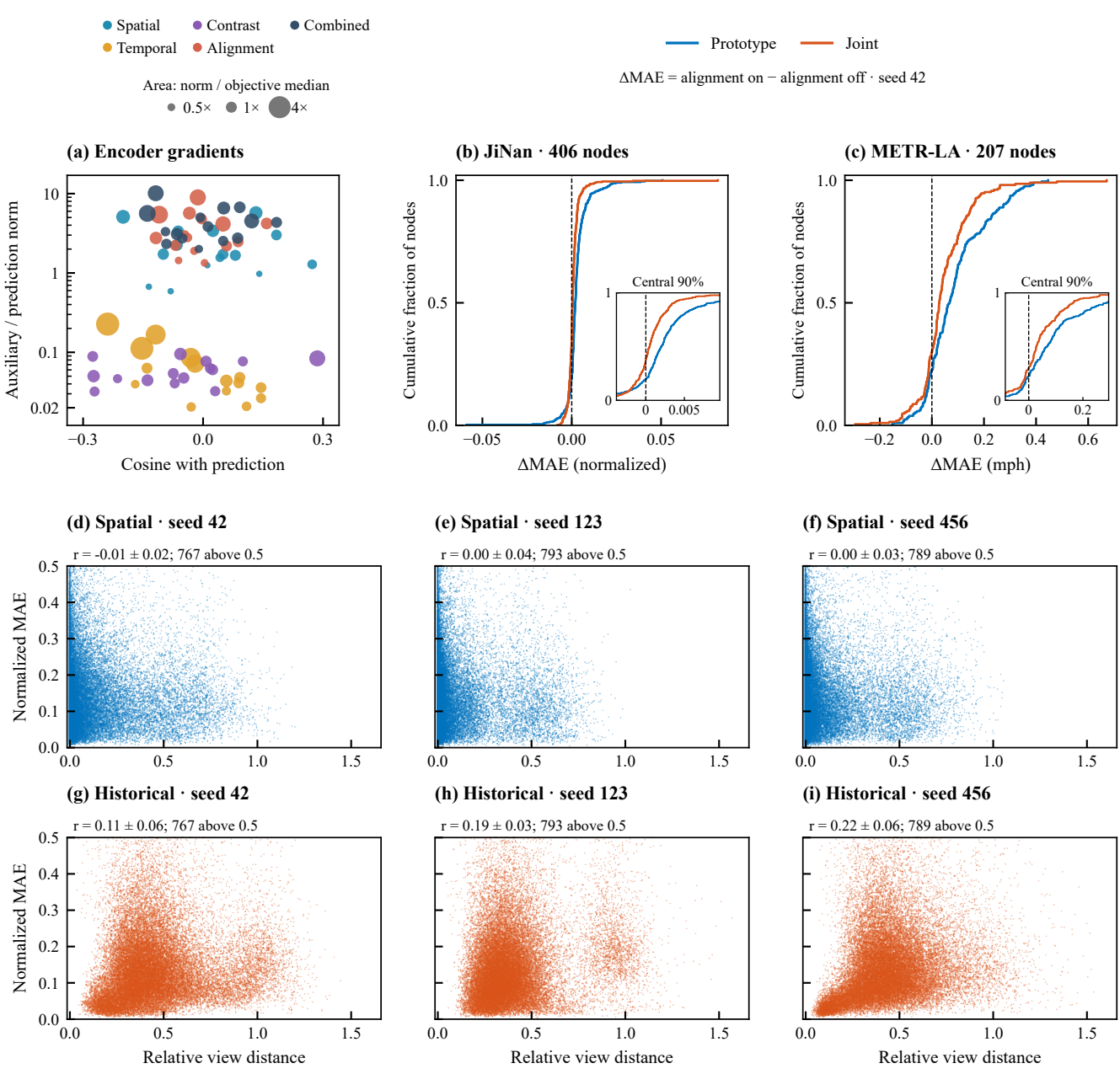

Spatial
Contrast
Combined
Temporal
Alignment
Area: norm / objective median
0.5×
1×
4×
Prototype
Joint
ΔMAE = alignment on − alignment off · seed 42
(a) Encoder gradients
Auxiliary / prediction norm
Cosine with prediction
(b) JiNan · 406 nodes
Cumulative fraction of nodes
ΔMAE (normalized)
Central 90%
(c) METR-LA · 207 nodes
Cumulative fraction of nodes
ΔMAE (mph)
Central 90%
(d) Spatial · seed 42
r = -0.01 ± 0.02; 767 above 0.5
(e) Spatial · seed 123
r = 0.00 ± 0.04; 793 above 0.5
(f) Spatial · seed 456
r = 0.00 ± 0.03; 789 above 0.5
Normalized MAE
(g) Historical · seed 42
r = 0.11 ± 0.06; 767 above 0.5
(h) Historical · seed 123
r = 0.19 ± 0.03; 793 above 0.5
(i) Historical · seed 456
r = 0.22 ± 0.06; 789 above 0.5
Normalized MAE
Relative view distance
Relative view distance
Relative view distance